SPL1 Project Proposal Form, 2026
Institute of Information Technology (IIT)
University of Dhaka

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Project Title: Face and Non-Face Identification Based on Partial Image Regions Using Different Method and Different Classifier .

Project Description: The proposed project aims to develop a C++-based system for identifying whether a JPG image is a Face or Non-Face based on partial image regions. Instead of analyzing only the complete image, the proposed system will divide the input image into multiple smaller regions or tiles and analyze the visual information contained in these local regions. This approach will allow the project to investigate how partial image information can be used for reliable Face/Non-Face identification.

The system will start with a dataset containing Face and Non-Face JPG images. For an input image, **image tiling** will first divide the image into several smaller regions. Features will then be extracted from each tile using a permitted external feature-extraction tool. The extracted information will be represented as numerical feature vectors. These feature vectors will be processed using **Max Pooling, Mean Pooling, and Attention Pooling** to create useful feature representations from the local image regions.

The processed feature representation will be given separately to two classifiers: **Random Forest and AdaBoost**. Random Forest will consist of multiple decision trees, where **Information Gain and Hellinger Distance** will be investigated as tree-splitting approaches. Information Gain evaluates how well a feature separates the classes, while Hellinger Distance measures the difference between class distributions. AdaBoost will be implemented independently as a boosting-based classifier using multiple weak learners that are trained sequentially.

After classification, the results from the relevant local regions or classification stages will be **aggregated** to produce an overall image-level prediction. The aggregation process will determine whether the combined evidence indicates a Face or Non-Face. The project will also allow comparison of different pooling techniques and the two separate classifiers to examine how they affect the final classification result.

The main focus of the project is the **from-scratch implementation of the major algorithms using C++**. Image tiling, pooling, attention processing, decision-tree operations, Information Gain, Hellinger Distance, Random Forest, AdaBoost, and prediction aggregation will be implemented without using OpenCV or general-purpose machine-learning libraries. A permitted external tool will be used only for the feature-extraction stage. The expected outcome is a working system that can classify JPG images as Face or Non-Face while demonstrating the application of local image analysis and ensemble learning techniques.

Languages or Tools to be used: C++

Supervisor's Name: Prof. Mohammad Shoyaib

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Midterm Presentation Feedback:

Before the Final Presentation:

I confirm that the project is complete and I am forwarding it for final presentation.

Signature of the supervisor: _____

Date: _____

